MELAKHESSOU YOUCEF

Seeking a 4 to 6-month internship starting March 29, 2027, in the field of Data, as a Data Analyst, Data Scientist, or on Business Intelligence (BI) oriented assignments.

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Scan to view my portfolio

PROFILE

Currently a Master's student in Decision Information Systems at the University of Lorraine, I am particularly interested in data analysis and data valorization. My academic background has enabled me to develop skills in Python, SQL, databases, and artificial intelligence through several projects involving data preparation, analysis, and exploitation. Rigorous and analytical, I am currently looking for an internship in the field of Data, in order to apply my skills and contribute to data-oriented and decision-support projects.

RESEARCH INTERNSHIP

LCOMS Laboratory – University of Lorraine **April-June 2025**

- Comparative analysis of community detection methods on graphs based on scientific papers and real and synthetic datasets.
- Construction of a taxonomy matrix to compare the performance, accuracy, scalability, and complexity of the approaches studied.
- Design of new local propagation strategies based on the selection of central nodes, multi-label propagation, and community core extraction.
- Study of Deep Learning approaches for graphs (GCN) and their integration with propagation methods.
- Analysis and visualization of results using Python / Jupyter Notebook, using metrics such as NMI and ARI and a computing environment with Spark/GPU.

ACADEMIC PROJECTS

Supervised Machine Learning Project **November-December 2024**

- Preparation, cleaning, and correction of a dataset from a database before use.
- Development of a supervised classification model to predict smoking status based on family and financial variables.
- Training of the model using the prepared data.
- Evaluation of the model's predictive capabilities on new data.

Oracle Database Design and Management Project **February-March 2025**

- Design of a relational database under Oracle for the management of projects, employees, equipment, and costs.
- Implementation of business rules and integrity constraints in PL/SQL to ensure data consistency.
- Development of PL/SQL triggers and procedures to automate controls and processing during INSERT, UPDATE, and DELETE operations.
- Automation of employee costs, equipment costs, and monthly cost calculations, as well as the monitoring of usage hours.
- Implementation of budget alerts in case of a project budget overrun.

Research Initiation Project **December-May 2026**

- Design and implementation in Java of three approaches for solving the Le Compte est Bon game: DFS with pruning, simulated annealing, and a hybrid approach.
- Design of a Chain of Responsibility architecture to modularize pruning techniques and analyze their impact on the search space.
- Experimental study and benchmark of the different approaches: execution time, number of explored nodes, and success rate.
- Development of a hybrid approach combining Simulated Annealing and DFS, reducing the search space from 21 trillion to 12,317 operations for n=12, in less than one second.

Artificial Intelligence Project **January-March 2026**

- Design of a hybrid game-playing agent combining supervised learning, heuristics, and MinMax search with Alpha-Beta pruning.
- Development and training of an MLP neural network on 303 expert decisions to learn move choices.
- Preparation and normalization of input data, followed by evaluation of the model's influence on decision-making.
- Implementation of a weighted decision function combining the outputs of the MLP, heuristic, and MinMax.
- Experimental testing and tuning of hyperparameters/coefficients to improve the bot's stability and performance

EDUCATION

Currently	**Master's Degree in Decision Information Systems - University of Lorraine - UFR MIM Metz**
2025-2026	**Master 1 in Computer Science - University of Lorraine - UFR MIM Metz**
2024-2025	**Bachelor's Degree, Year 3 in Computer Science - University of Lorraine - UFR MIM Metz**
2023-2024	**Bachelor's Degree, Year 2 in Computer Science - University of Lorraine - UFR MIM Metz**
2022-2023	**Bachelor's Degree, Year 1 in Computer Science-Mathematics-Mechanics-Physics-ACCES SANTE - Aix Marseille Université**
2021-2022	**Bac scientifique spécialité science de la vie et de la terre - Lycée les frères Lamrani Algérie**

SKILLS

Soft Skills

- **Organizational skills**
- **Teamwork**
- **Analytical thinking**

Programming Languages

C	: Good level	**SQL**	: Proficient
C++	: Good level	**JavaScript**	: Good level
Java	: Proficient	**PHP**	: Proficient
Python	: Good level	**TypeScript**	: Good level

Tools

- **Visual Studio**
- **PowerBI**
- **IntelliJ IDEA**
- **GitHub**

LANGUAGES

French : **Fluent**

Arabic : **Native language**

English : **Intermediate (B2)**

INTERESTS

- **Automotive**
- **Travel**
- **Gaming**
- **Swimming**